the first time that the stars in disc were indeed younger than the stars in the bulge.

One of the instruments that gave astronomers unprecedented insight into our galactic history was the Gaia telescope. The instrument was used to figure out the precise brightness, position and distance of a million stars in the Milky Way 6,500 light years around the Sun. The total number of stars in the Milky Way is estimated to be around a hundred thousand million stars. The data from the Gaia instrument allowed researchers to unravel a colossal feeding event from the Milky Way's dark, cannibalistic past.

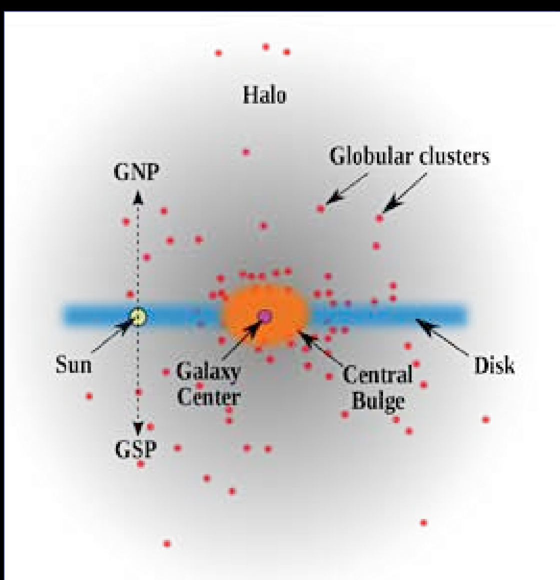
This was because the variation in the age and metallicity of the stars in the halo region had a much wider range than expected. One of the things to understand here is that metallicity, age and the resulting colours of the stars are all relative. The population II stars found within the halo are older and poorer in metals than the population II stars found in the bulge. Within the Halo, the Gaia researchers honed in on some stars that were bluer than the rest, and isolated them as having originated from a galaxy that had been swallowed up by the Milky Way in the distant past. While the age of the stars in both the components that made up the galaxy were equally old, the stars within Gaia-Enceladus, the dwarf galaxy that the Milky Way swallowed up, had stars that were poorer in metals than the Milky Way progenitor galaxy. The progenitor galaxy was four times the size of Gaia-Enceladus. The event occurred thirteen thousand million years ago, and observations of other spiral galaxies have indicated that such events are fairly common in the past of a galaxy, and is very likely to be related to why the universe keeps making spiral galaxies.

Credit: Gabriel Perez Diaz, SMM (IAC)

The collision with Gaia-Enceladus

Researchers from Caltech have studied ancient star clusters, as well as the star clusters in our own galaxy to try and understand which of the stars originated in our own galaxy, and which originated outside. By looking specifically at the star clusters, the researchers managed to identify at least five satellite galaxies that the Milky Way has swallowed up in the distant past. While these galaxies have been assimilated into the Milky Way, and their forms have been destroyed long ago, small clusters of stars from these original galaxies have survived and remained together, allowing astronomers to reverse engineer the sequence of events that led to the formation of the Milky Way. A supercomputer simulation shows an early time period when a few dwarf galaxies merged together to form the main body of the Milky Way, which began spinning increasingly rapidly with the acquired mass. With an increasing number of collisions and brushes, the galaxy grew as it acquired stars and matter over a series of interactions.

In 2020 new research indicted how the dark matter in neighbouring galaxies played a much bigger role in shaping the Milky Way than scientists previously suspected. The Milky Way is headed towards a point where the Large Magellanic Cloud was in the past. This is because the dark matter halo of the LMC is pulling the disc of the Milky Way towards the constellation of Pegasus at a rate of 32 km/s, and at the same time powering its movement, fleeing from the Mlky Way at 370 km/s. What happened to the Milky Way is very much a subject of ongoing research. While our position inside the galaxy gives us the best possible close up views, we still do not understand a lot about the galaxy such as its total luminosity, its radial structure or its rotational curve as easily as we can in galaxies that we can observe in totality using our astronomical instruments. d



Credit: Gabriel Perez Diaz, SMM (IAC)

Schematic of the Milky Way